

Table Tennis Trajectory

Ruel Sinha, Owen Marsden

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0.1 Simulated Equation

Given the first two data points, we can approximate the initial velocity using a secant line. Time is divided by 8, because of slow motion. Points: (0,0.0003499487, -0.0008807043), (0.005,-0.05580366,0.008784190)
We will find the velocity of the y and x components separately.

$$m_x = \frac{x_2 - x_1}{t_2 - t_1} = \frac{|-0.0558036600 - 0.0003499487|}{0.005} = 11.23072174$$

$$m_y = \frac{y_2 - y_1}{t_2 - t_1} = \frac{-0.008784190 - 0.0008807043}{0.005} = 1.93297886$$

The velocity is equal to the magnitude of the vector with respective x and y components, so

$$m = \sqrt{11.23072174^2 + 1.93297886^2} = 11.3958553$$

The angle exists such that $\tan\theta = \frac{m_y}{m_x}$

$$\theta = \arctan\left(\frac{1.93297886}{11.23072174}\right) = 0.1704453571\text{rad}$$

since air resistance isn't real, we can assume that $x \propto t$ Therefore, we can determine the relationship that

$$x = 11.23072174t, \quad t = 0.08904147241x$$

since air resistance also isn't real in the y dimension, the only force acting on the ball is gravity. This force, being the second derivative of position, decreases the velocity at a rate of $9.81m/s^2$

$$\therefore a_y = -9.81$$

We will now integrate to find the position.

$$v_y(t) = \int a_y(t)dt = \int -9.81dt = -9.81t + v_0$$

v_{oy} is simply the initial value we found before, so

$$v_y(t) = -9.81t + 1.93297886$$

now to find position, we integrate again

$$s_y(t) = \int v_y(t)dt = \int -9.81t + 1.93297886dt = -4.905t^2 + 1.93297886t + s_0$$

we know that the ball started at $y = -0.0008807043$, so $s_0 = -0.0008807043$ To establish an explicit equation of x and y, we can recall our finding that $t = 0.08904147241x$ substituting it in gives us

$$y(x) = -4.905(0.08904147241x)^2 + 1.93297886(0.08904147241x) + -0.0008807043$$

$$y(x) = -0.03888872258x^2 + 0.1721152838x - 0.0008807043$$

0.2 Real Equation

let $s(t)$ = the parametric function for the ball's position for a time t .

$$s(t) = x(t)\mathbf{i} + y(t)\mathbf{j}, \quad ms''(t) = F(t)$$

$$F(t) = F_g + F_D + F_M$$

$$F_g = -mg\mathbf{j} = \text{force of gravity}$$

$$F_D = -k_D(x'\mathbf{i} + y'\mathbf{j}) = \text{air resistance}$$

$$F_M = C_0(-\omega\mathbf{k} \times (x'\mathbf{i} + y'\mathbf{j})) = k_L(y'\mathbf{i} - x'\mathbf{j}) = \text{lift force (caused by constant angular velocity } \omega)$$

where k_D , C_0 , k_L , and g are positive constants. $C_0\omega = k_L$, $g = 9.81m/s^2$

$$F(t) = ms'' = -mg\mathbf{j} - k_D(x'\mathbf{i} + y'\mathbf{j}) + k_L(y'\mathbf{i} - x'\mathbf{j})$$

$$mx''\mathbf{i} + my''\mathbf{j} = (-k_Dx' + k_Ly')\mathbf{i} - (k_Lx' + k_Dy' + mg)\mathbf{j}$$

$$\text{let } u = x', v = y'$$

$$mu' = -k_Du + k_Lv$$

$$k_Lv = mu' + k_Du$$

$$v = \frac{m}{k_L}u' + \frac{k_D}{k_L}u$$

$$v' = \frac{m}{k_L}u'' + \frac{k_D}{k_L}u'$$

$$mv' = -k_Lu - k_Dv - mg$$

$$m\left(\frac{m}{k_L}u'' + \frac{k_D}{k_L}u'\right) = -k_Lu - k_D\left(\frac{m}{k_L}u' + \frac{k_D}{k_L}u\right) - mg$$

$$\frac{m^2}{k_L}u'' + \frac{mk_D}{k_L}u' = -k_Lu - \frac{mk_D}{k_L}u' - \frac{k_D^2}{k_L}u - mg$$

$$\frac{m^2}{k_L}u'' + \frac{2mk_D}{k_L}u' + \left(k_L + \frac{k_D^2}{k_L}\right)u + mg = 0$$

$$\text{let } \frac{m^2}{k_L} = a, \quad \frac{2mk_D}{k_L} = b, \quad \left(k_L + \frac{k_D^2}{k_L}\right) = c, \quad mg = d$$

$$au'' + bu' + cu + d = 0$$

$$\text{first, let's solve } au_c'' + bu_c' + cu_c = 0$$

$$\text{let } u_c = e^{ht}, \quad u_c' = he^{ht}, \quad u_c'' = h^2 e^{ht}$$

$$ah^2 e^{ht} + bhe^{ht} + ce^{ht} = 0$$

$$(ah^2 + bh + c) e^{ht} = 0$$

$$ah^2 + bh + c = 0 \text{ because } e^{ht} \neq 0$$

$$h = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$h = \frac{-\frac{2mk_D}{k_L} \pm \sqrt{\left(\frac{2mk_D}{k_L}\right)^2 - 4\frac{m^2}{k_L}\left(k_L + \frac{k_D^2}{k_L}\right)}}{2\frac{m^2}{k_L}}$$

$$h = -\frac{k_D}{m} \pm \sqrt{\left(\frac{k_D}{m}\right)^2 - \frac{k_L}{m^2}\left(k_L + \frac{k_D^2}{k_L}\right)}$$

$$h = -\frac{k_D}{m} \pm \sqrt{\frac{k_D^2}{m^2} - \frac{k_L^2}{m^2} - \frac{k_D^2}{m^2}}$$

$$h = -\frac{k_D}{m} \pm \sqrt{-\frac{k_L^2}{m^2}} = -\frac{k_D}{m} \pm i\frac{k_L}{m}$$

Arbitrarily set $u_c = e^{\left(-\frac{k_D}{m} + i\frac{k_L}{m}\right)t} + e^{\left(-\frac{k_D}{m} - i\frac{k_L}{m}\right)t}$ as an example.

$$\text{let } \frac{k_D}{m} = p, \quad \frac{k_L}{m} = q$$

$$u_c = e^{-pt} (e^{iqt} + e^{-iqt})$$

$$\text{Imaginary definition of cosine: } \cos(z) = \frac{e^{iz} + e^{-iz}}{2}$$

$$\therefore u_c = 2e^{-pt} \cos(qt)$$

We can write the general solution as $\alpha e^{-pt} \cos(qt) + \beta e^{-pt} \sin(qt)$

since their Wronskian is not equal to zero (α and β are constants).

We can guess a particular solution $u_0 = u_c - \frac{d}{c}$

$$\text{for } au_0'' + bu_0' + cu_0 + d = 0$$

$$au_c'' + bu_c' + c \left(u_c - \frac{d}{c} \right) + d = 0$$

$$(au_c'' + bu_c' + cu_c) - d + d = 0$$

Theorem for general solution: $u = u_0 + u_c$

$$u = \alpha e^{-pt} \cos(qt) + \beta e^{-pt} \sin(qt) - \frac{d}{c}$$

$$u = e^{-pt} (\alpha \cos(qt) + \beta \sin(qt)) - \frac{d}{c}$$

$$u' = -pe^{-pt} (\alpha \cos(qt) + \beta \sin(qt)) + e^{-pt} (\beta q \cos(qt) - \alpha q \sin(qt))$$

$$u' = e^{-pt} (-\alpha p \cos(qt) - \beta p \sin(qt) + \beta q \cos(qt) - \alpha q \sin(qt))$$

$$u = e^{-pt} (\alpha \cos(qt) + \beta \sin(qt)) - \frac{mg}{k_L + \frac{k_D^2}{k_L}}$$

$$u' = e^{-pt} \left(-\frac{k_D}{m} \alpha \cos(qt) - \frac{k_D}{m} \beta \sin(qt) + \frac{k_L}{m} \beta \cos(qt) - \frac{k_L}{m} \alpha \sin(qt) \right)$$

$$v = \frac{m}{k_L} u' + \frac{k_D}{k_L} u$$

$$v = \frac{m}{k_L} e^{-pt} \left(-\frac{k_D}{m} \alpha \cos(qt) - \frac{k_D}{m} \beta \sin(qt) + \frac{k_L}{m} \beta \cos(qt) - \frac{k_L}{m} \alpha \sin(qt) \right) + \frac{k_D}{k_L} \left(e^{-pt} (\alpha \cos(qt) + \beta \sin(qt)) - \frac{mg}{k_L + \frac{k_D^2}{k_L}} \right)$$

$$v = e^{-pt} \left(-\frac{k_D}{k_L} \alpha \cos(qt) - \frac{k_D}{k_L} \beta \sin(qt) + \beta \cos(qt) - \alpha \sin(qt) + \frac{k_D}{k_L} \alpha \cos(qt) + \frac{k_D}{k_L} \beta \sin(qt) \right) - \frac{mgk_D}{k_L^2 + k_D^2}$$

$$v = e^{-pt} (\beta \cos(qt) - \alpha \sin(qt)) - \frac{mgk_D}{k_L^2 + k_D^2}$$

$$x(t) = \int u \, dt$$

$$x(t) = \int e^{-pt} (\alpha \cos(qt) + \beta \sin(qt)) - \frac{mgk_L}{k_L^2 + k_D^2} dt$$

$$x(t) = \alpha \int e^{-pt} \cos(qt) \, dt + \beta \int e^{-pt} \sin(qt) \, dt - \int \frac{mgk_L}{k_L^2 + k_D^2} dt$$

$$x(t) = \alpha \int e^{-pt} \cos(qt) \, dt + \beta \int e^{-pt} \sin(qt) \, dt - \frac{mgk_L}{k_L^2 + k_D^2} t$$

$$\int e^{-pt} \cos(qt) \, dt \text{ using DI method:}$$

	D	I
+	e^{-pt}	$\cos(qt)$
-	$-pe^{-pt}$	$\frac{1}{q} \sin(qt)$
+	$p^2 e^{-pt}$	$-\frac{1}{q^2} \cos(qt)$

$$\int e^{-pt} \cos(qt) \, dt = \frac{1}{q} e^{-pt} \sin(qt) - \frac{p}{q^2} e^{-pt} \cos(qt) - \frac{p^2}{q^2} \int e^{-pt} \cos(qt) \, dt$$

$$\left(1 + \frac{p^2}{q^2}\right) \int e^{-pt} \cos(qt) \, dt = \frac{1}{q} e^{-pt} \sin(qt) - \frac{p}{q^2} e^{-pt} \cos(qt)$$

$$\int e^{-pt} \cos(qt) \, dt = \frac{\frac{1}{q} e^{-pt} \sin(qt) - \frac{p}{q^2} e^{-pt} \cos(qt)}{1 + \frac{p^2}{q^2}}$$

$$\int e^{-pt} \cos(qt) \, dt = \frac{qe^{-pt} \sin(qt) - pe^{-pt} \cos(qt)}{q^2 + p^2}$$

$$\int e^{-pt} \cos(qt) \, dt = \frac{e^{-pt}}{q^2 + p^2} (q \sin(qt) - p \cos(qt))$$

$$\int e^{-pt} \sin(qt) \, dt \text{ using DI method:}$$

	D	I
+	e^{-pt}	$\sin(qt)$
-	$-pe^{-pt}$	$-\frac{1}{q} \cos(qt)$
+	$p^2 e^{-pt}$	$-\frac{1}{q^2} \sin(qt)$

$$\int e^{-pt} \cos(qt) dt = -\frac{1}{q} e^{-pt} \cos(qt) - \frac{p}{q^2} e^{-pt} \sin(qt) - \frac{p^2}{q^2} \int e^{-pt} \sin(qt) dt$$

$$\left(1 + \frac{p^2}{q^2}\right) \int e^{-pt} \sin(qt) dt = -\frac{1}{q} e^{-pt} \cos(qt) - \frac{p}{q^2} e^{-pt} \sin(qt)$$

$$\int e^{-pt} \sin(qt) dt = \frac{-\frac{1}{q} e^{-pt} \cos(qt) - \frac{p}{q^2} e^{-pt} \sin(qt)}{1 + \frac{p^2}{q^2}}$$

$$\int e^{-pt} \sin(qt) dt = \frac{-qe^{-pt} \cos(qt) - pe^{-pt} \sin(qt)}{q^2 + p^2}$$

$$\int e^{-pt} \sin(qt) dt = \frac{-e^{-pt}}{q^2 + p^2} (q \cos(qt) + p \sin(qt))$$

$$x(t) = \alpha \frac{e^{-pt}}{q^2 + p^2} (q \sin(qt) - p \cos(qt)) - \beta \frac{e^{-pt}}{q^2 + p^2} (q \cos(qt) + p \sin(qt)) - \frac{mgk_L}{k_L^2 + k_D^2} t + C_x$$

$$x(t) = \frac{e^{-pt}}{\frac{k_L^2}{m^2} + \frac{k_D^2}{m^2}} \alpha (q \sin(qt) - p \cos(qt)) - \frac{e^{-pt}}{\frac{k_L^2}{m^2} + \frac{k_D^2}{m^2}} \beta (q \cos(qt) + p \sin(qt)) - \frac{mgk_L}{k_L^2 + k_D^2} t + C_x$$

$$x(t) = \frac{e^{-pt} m^2}{k_L^2 + k_D^2} \alpha \left(\frac{k_L}{m} \sin(qt) - \frac{k_D}{m} \cos(qt) \right) - \frac{e^{-pt} m^2}{k_L^2 + k_D^2} \beta \left(\frac{k_L}{m} \cos(qt) + \frac{k_D}{m} \sin(qt) \right) - \frac{mgk_L}{k_L^2 + k_D^2} t + C_x$$

$$x(t) = \frac{me^{-\frac{k_D}{m}t}}{k_L^2 + k_D^2} \left(k_L \alpha \sin\left(\frac{k_L}{m}t\right) - k_D \alpha \cos\left(\frac{k_L}{m}t\right) - k_L \beta \cos\left(\frac{k_L}{m}t\right) - k_D \beta \sin\left(\frac{k_L}{m}t\right) - gk_L t e^{\frac{k_D}{m}t} \right) + C_x$$

$$y(t) = \int \mathbf{v} dt$$

$$y(t) = \int e^{-pt} (\beta \cos(qt) - \alpha \sin(qt)) - \frac{mgk_D}{k_L^2 + k_D^2} dt$$

This expression takes the same form as $x(t) = \int u dt$ with a few constants swapped around.

$$y(t) = \frac{me^{-\frac{k_D}{m}t}}{k_L^2 + k_D^2} \left(k_L \beta \sin\left(\frac{k_L}{m}t\right) - k_D \beta \cos\left(\frac{k_L}{m}t\right) + k_L \alpha \cos\left(\frac{k_L}{m}t\right) + k_D \alpha \sin\left(\frac{k_L}{m}t\right) - gk_D t e^{\frac{k_D}{m}t} \right) + C_y$$

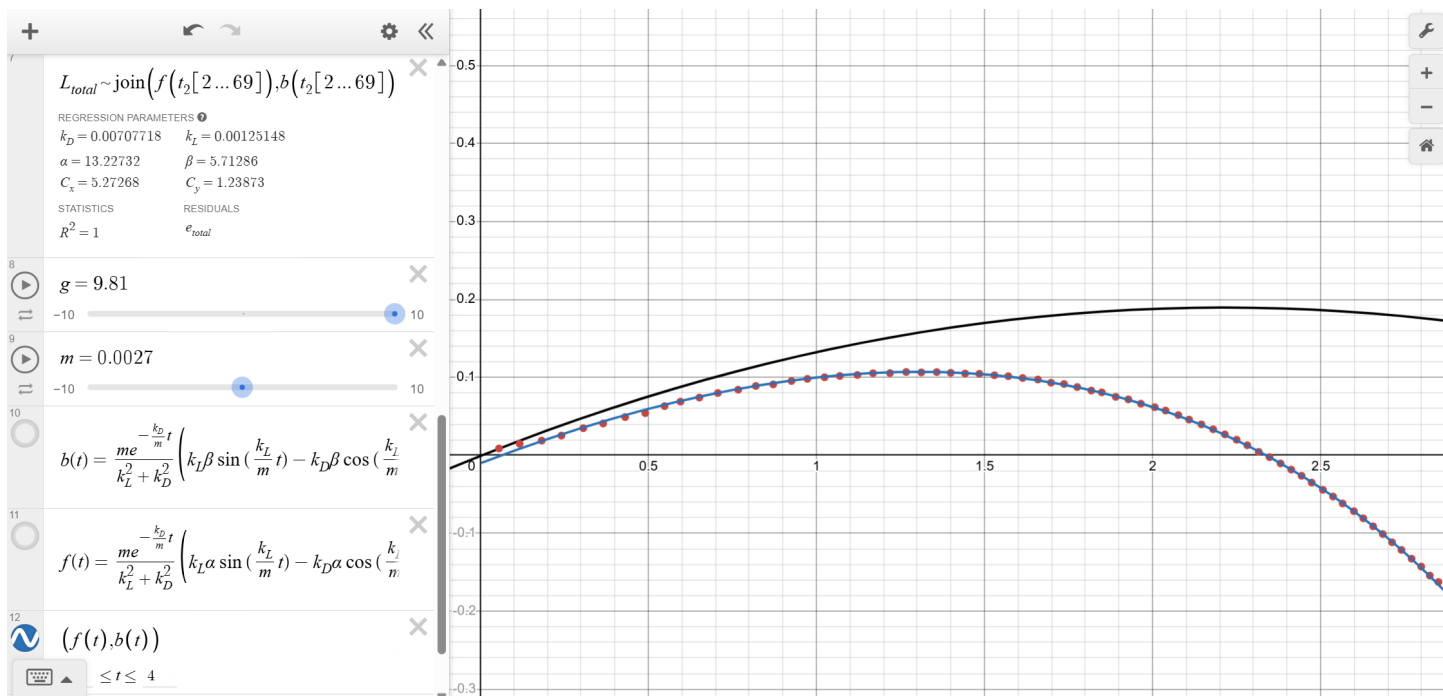


Figure 1: Quadratic and Non-quadratic models alongside data

References:

<https://tutorial.math.lamar.edu/Classes/DE/IntroSecondOrder.aspx>

<https://www.desmos.com/calculator>